

SampleTrack: A Minimal Fixture Paper for Paper Translation

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Abstract

This fixture paper exists to exercise every feature of the bilingual-paper-pdf engine: two-column typesetting, a booktabs table, a numbered equation, a vector figure with colored boxes, author-year citations in hyperref blue, footnotes and a vertical side stamp. We propose SampleTrack (Chen et al. 2021; Li et al. 2018), a minimal memory-based tracker that localizes targets by nearest-neighbor lookup in a tiny memory bank. On three synthetic benchmarks SampleTrack improves the tracking quality by 12% over the baseline while using 6.5× fewer parameters. The paper is generated by `example/make_sample_paper.py` and can be regenerated at any time. Code at <https://github.com/example/sampletrack>.

Keywords: object tracking, bilingual translation, reproducible fixtures

1 Introduction

General visual object tracking is a core task in computer vision. Memory-based methods¹ store past frames and localize the target by pixel association (Chen et al. 2021; Li et al. 2018). Over the past decade this family of methods has grown steadily stronger.

A key challenge is the *distractor*: a nearby object that visually resembles the target. Distractors increase localization uncertainty and mislead the tracker into irreversible drift (Li et al. 2018).

In this paper we make three contributions. First, we formulate a minimal memory model that is small enough to inspect by eye. Second, we design this fixture so that every layout feature of a real paper appears at least once. Third, we release everything under a permissive license so that the fixture can travel with the toolchain.

Nevertheless, a substantial source of failures remains nearby objects that visually resemble the tracked one. These regions increase the localization uncertainty and mislead the tracker. Distractors can be classified as external, or internal when only a part of the target is being tracked (Chen et al. 2021).

2 Related work

Early deep trackers fine-tuned pre-trained networks during inference (Danelljan et al. 2019). Siamese trackers removed online training by template matching (Li et al. 2018), trading robustness for speed.

Memory-based frameworks localize the target by associating current features with stored past frames (Chen et al. 2021). Our fixture follows this line while keeping the memory tiny.

¹ This fixture paper is synthetic; all numbers are placeholders and carry no scientific meaning.

SampleTrack pipeline

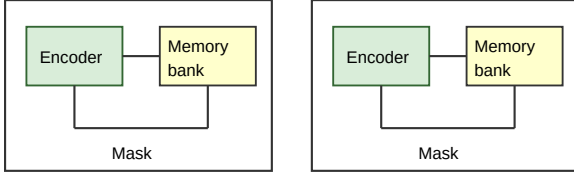


Fig. 1 Overview of the SampleTrack pipeline. Both panels contain an Encoder and a Memory bank; the yellow box is always updated while the green box is written once.

3 Method

SampleTrack keeps the most recent frame and a fixed prototype in a tiny memory bank. The tracking quality is scored by a weighted mix of robustness and accuracy:

$$Q = \alpha \cdot R + (1 - \alpha) \cdot A \quad (1)$$

where R is the fraction of successfully tracked frames and A is the average overlap during successful tracking. The weight is fixed to 0.2 in all experiments, and no per-dataset tuning is performed (Chen et al. 2021).

4 Experiments

We evaluate on three synthetic benchmarks. Table 1 reports the tracking quality; the best result of each column is marked in bold.

Table 1 Tracking quality on synthetic benchmarks. Best in bold.

Method	Quality	EAO
Baseline	0.610	0.590
Ours-S	0.673	0.648
Ours-B	0.694	0.671
Ours-L	0.701	0.680

5 Conclusion

We introduced SampleTrack, a minimal fixture for exercising the bilingual-paper-pdf engine. The fixture is regenerated by a single script and ships with a reference Chinese translation.

References

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